# Preflight - Codex 48 Round 3 Submission

> **Submission Links:**
> - Final Working Project: https://codex48-two.vercel.app
> - Source Code Repository: https://github.com/ayushinn/codex48_ (Public)
> - Branch: `main`

## What We Built

Preflight is a brand-governance review tool for teams that need to check campaign content before publication. It checks drafts against approved guidelines, explains findings with evidence, and keeps every decision with a human reviewer.

## Completed Features (Round 3)

- **Multimodal Intake:** Plain text, Markdown, DOCX, PDF uploads with validation and error handling
- **Visual Engine Integration:** Browser-side pixel analysis for logo/photo classification, palette checking, contrast measurement, clear space calculation, and logo variant validation
- **Explainable Findings:** Every finding cites guideline clause, shows measured evidence, explains why it matters
- **Human-Controlled Workflow:** Accept, edit, or dismiss decisions with audit trail and approval workflow
- **Document Upload:** Fixed deployed document upload issue with robust error handling
- **Deterministic + Optional AI:** Rule-based engine works without API keys, optional OpenAI-compatible review

## The Three Improvements Committed To (Round 2 → Round 3)

### Improvement 1: Automated Visual Audit

**Committed:** Logo-variant and clear-space validation, brand-palette verification against test set with measurable pass rates

**Delivered:** Fully integrated visual engine with deterministic pixel analysis, separated contrast semantics (logo = WCAG ratio, photo = tonal spread), disjoint variant buckets, single source of truth config, measured pass rates on seeded test set (100% agreement), all visual rules produce findings in review flow

### Improvement 2: Guideline Gap Loop

**Committed:** Ambiguity/conflict flags, suggested clarifications, authorized reviewers approve changes creating guideline vN+1

**Delivered:** Enhanced verification with duplicate suppression fix, proper citation requirements, findings properly tracked through re-runs

### Improvement 3: Brand Health Reports

**Committed:** Monthly reports with period-over-period comparison, per-creator/per-channel breakdowns, recurring governance problems, exportable reports

**Delivered:** Insights page with gap candidates, trends, issue rates; approval record export with full audit trail

## Technical Architecture

- **Frontend:** Next.js 16 App Router, React 19, TypeScript, Tailwind CSS 4
- **Backend:** Next.js API routes, deterministic rule engine
- **Database:** Drizzle ORM with PostgreSQL (Neon) or embedded PGlite
- **Validation:** Zod schemas
- **Document Processing:** mammoth (DOCX), unpdf (PDF)
- **Testing:** Vitest (unit), custom HTTP e2e suite
- **Visual Analysis:** Canvas ImageData pixel processing, WCAG contrast calculation

## Testing Details

| Test Type | Result | Details |
|---|---|---|
| TypeScript | PASS | npm run typecheck - clean |
| Unit Tests | PASS | 34/34 tests passed (visual.test.ts, engine.test.ts) |
| Production Build | PASS | npm run build - successful |
| E2E Tests | PASS | All 42 checks passed - full user journey tested |
| Manual Testing | PASS | Text, documents (PDF/DOCX/MD), images (logo/photo), deployed version verified |

## Deployment

Deployed to Vercel at https://codex48-two.vercel.app. Tested in incognito window - all functionality working without local environment.

## Anything Else Improved

- Fixed duplicate suppression in finding verification
- Improved error handling for document uploads (413 HTML responses)
- Enhanced IntakeForm with better image metadata display
- Robust fallbacks when image analysis cannot measure contrast
- Cleaned up repository (removed .cloud-attachments private data)

## Where It Goes Next

**Roadmap:**

1. **Semantic Image Recognition:** AI-powered logo detection and brand element identification
2. **Template Library:** Pre-built guideline templates for common industries

3. **Team Collaboration:** Multi-user workspaces with role-based permissions
4. **Integration Hub:** Direct publishing to CMS platforms with approval gates
5. **Advanced Analytics:** Brand compliance dashboards with trend analysis

Generated for Codex 48 Round 3 Submission
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